Problem Statement: Syllabus to Quiz Generator

The Challenge

Students and educators often struggle to convert large, unstructured syllabus content into meaningful assessments. Syllabi may include textbooks, lecture notes, PDFs, and slides spread across multiple sources, making it difficult to identify key concepts and generate relevant questions. Manual quiz creation is time-consuming, inconsistent in quality, and often lacks alignment with learning objectives or difficulty levels. Additionally, adapting quizzes for different learners and tracking progress adds further complexity.

The Objective

Build an agentic AI-powered Syllabus-to-Quiz Generator that acts as an intelligent assessment assistant. The solution should:

1. **Transform Syllabus into Structured Content** – Extract and organize topics, subtopics, and key concepts from syllabus documents and learning materials.
2. **Automate Quiz Generation** – Generate high-quality quizzes (MCQs, short answers, descriptive) aligned with difficulty levels and learning outcomes.
3. **Enable Adaptive Assessment** – Adjust question difficulty based on user performance and learning progress.
4. **Support Multimodal Inputs** – Accept PDFs, lecture notes, images, and text to generate quizzes from diverse educational content.
5. **Provide Feedback & Analytics** – Offer instant evaluation, explanations, and performance insights through interactive dashboards.
6. **Ensure Personalization** – Tailor quizzes based on student level, subject focus, and exam requirements.

This enables a smart, scalable, and efficient system that simplifies assessment creation while improving learning outcomes.